

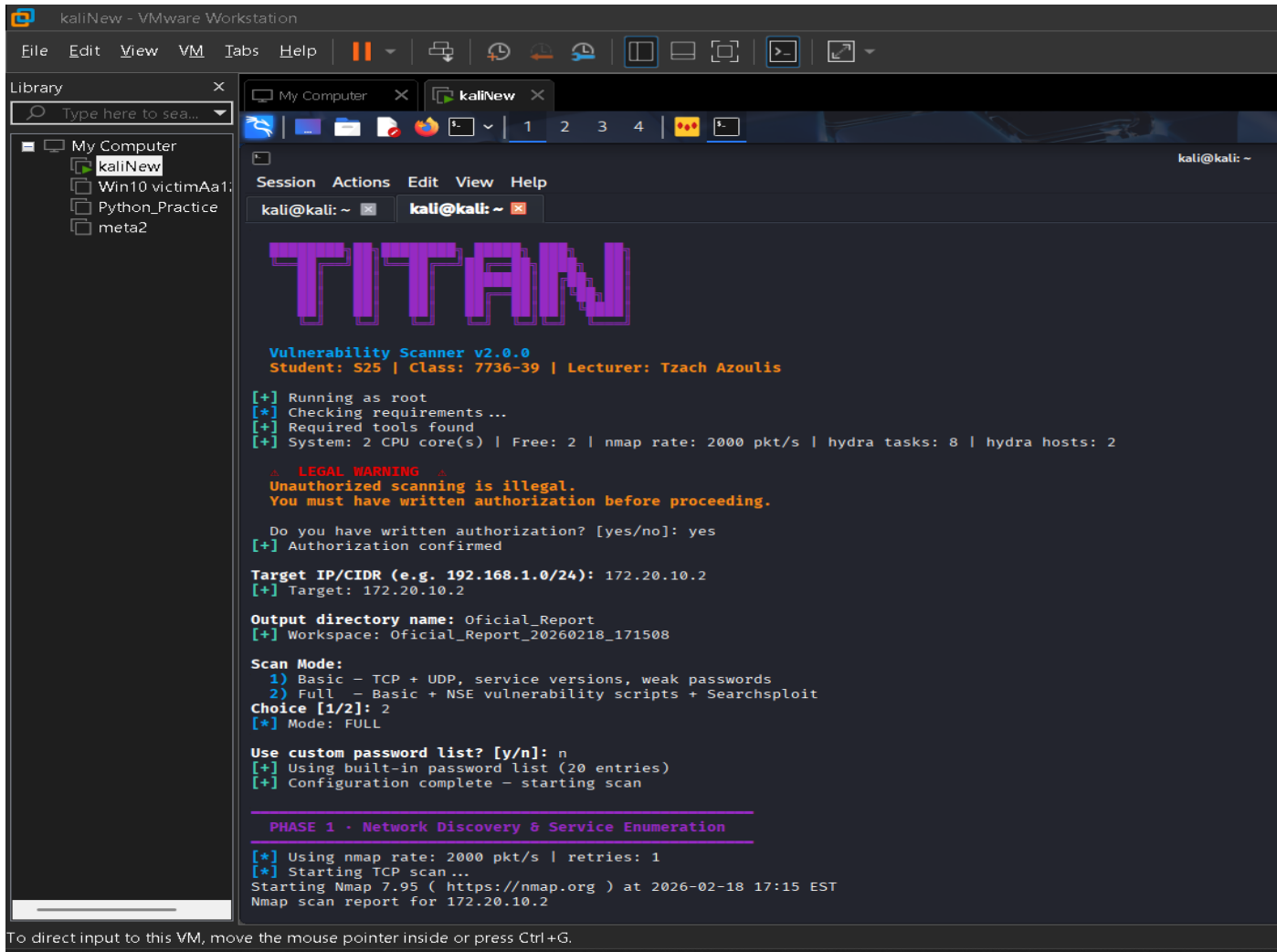
Project Documentation: TITAN Vulnerability Scanner

Program Code: ZX301

Unit: 7736-39 Student ID: S25 Lecturer: Tzach Azoulis Date: February 19, 2026

1. Initialization and Input Validation

The TITAN Vulnerability Scanner (v2.0.0) is designed to enforce strict procedural constraints before execution. The script begins by verifying root privileges to ensure full access to raw socket operations required by Nmap. Following this, the user is prompted to validate the target network using CIDR notation and select the output directory. To comply with the "Operation Order," the user must explicitly choose between "Basic" and "Full" scanning modes. The input validation logic ensures that invalid IP addresses or unauthorized execution attempts are rejected immediately.



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2. Network Discovery and Service Enumeration

Upon successful initialization, the scanner executes the network discovery phase. In accordance with Criteria 1.3, the script performs a comprehensive scan of both TCP and UDP protocols. The TCP scan utilizes SYN packets for stealth, while the UDP scan targets the top 100 most common ports to optimize performance without sacrificing visibility. This phase also includes service version detection (-sV) and Operating System fingerprinting (-O) to build a complete profile of the target host.

(Place Screenshot 2 here: Showing Phase 1 start and the TCP scan results with open ports like 135, 139, 445, 5432)

```
kaliNew - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
kaliNew
Win10 victimAa1
Python_Practice
meta2

Session Actions Edit View Help
kali@kali: ~ kali@kali: ~

PHASE 1 - Network Discovery & Service Enumeration

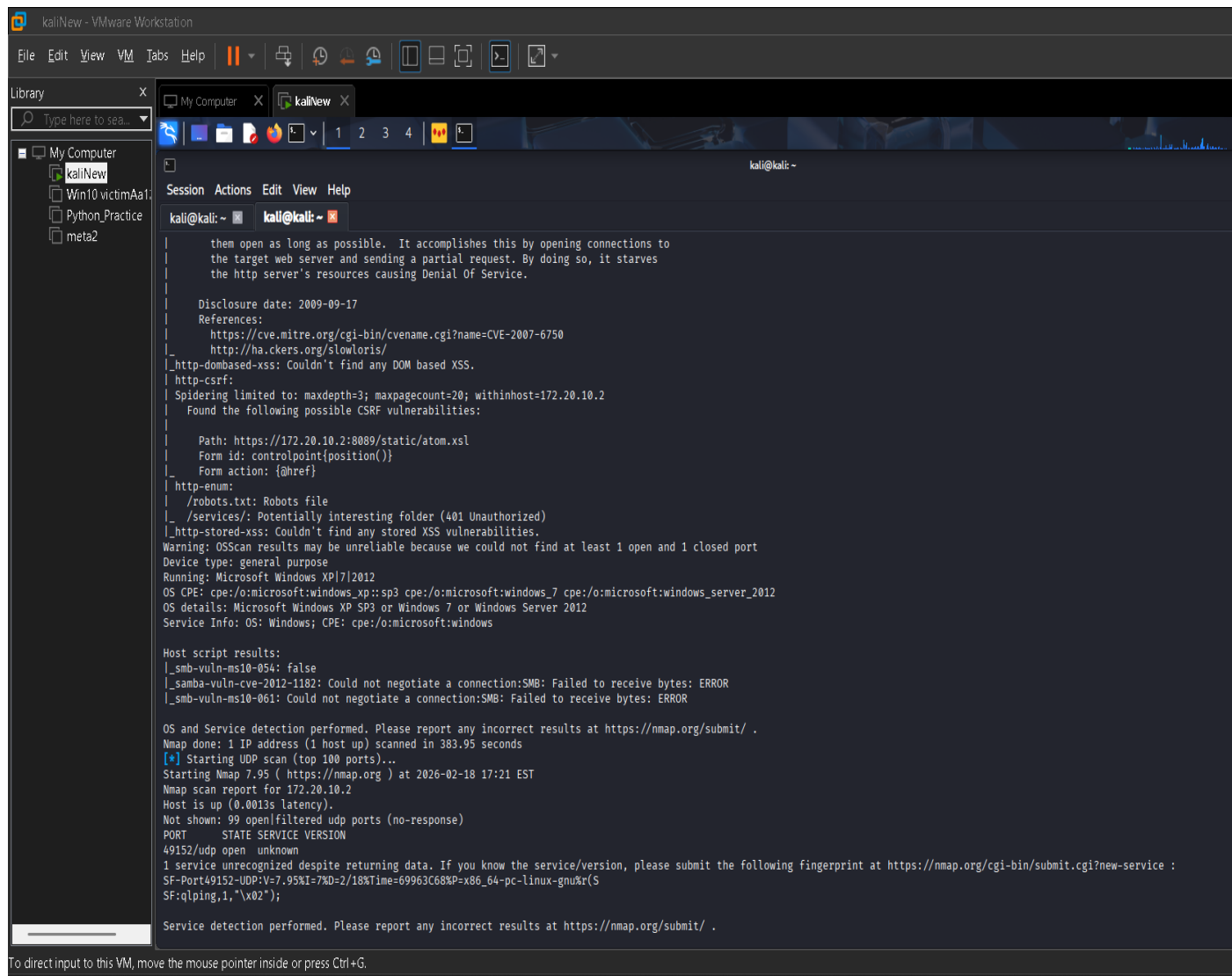
[*] Using nmap rate: 2000 pkt/s | retries: 1
[*] Starting TCP scan ...
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-18 17:15 EST
Nmap scan report for 172.20.10.2
Host is up (0.0014s latency).
Not shown: 993 filtered tcp ports (no-response)
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT      STATE SERVICE        VERSION
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
|_ssl-ccs-injection: No reply from server (TIMEOUT)
5432/tcp  open  postgresql     PostgreSQL DB 9.6.0 or later
8001/tcp  open  http           Splunkd httpd
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-slowloris-check:
|_  VULNERABLE:
|_    Slowloris DOS attack
|_      State: LIKELY VULNERABLE
|_      IDs: CVE:CVE-2007-6750
|_      Slowloris tries to keep many connections to the target web server open and hold
|_      them open as long as possible. It accomplishes this by opening connections to
|_      the target web server and sending a partial request. By doing so, it starves
|_      the http server's resources causing Denial Of Service.
|_
|_  Disclosure date: 2009-09-17
|_  References:
|_    https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|_    https://ha.ckers.org/slowloris/
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
|_http-server-header: Splunkd
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-enum:
|_  /robots.txt: Robots file
8089/tcp  open  ssl/http       Splunkd httpd
|_http-server-header: Splunkd
|_http-slowloris-check:
|_  VULNERABLE:
|_    Slowloris DOS attack
|_      State: LIKELY VULNERABLE
|_      IDs: CVE:CVE-2007-6750
|_      Slowloris tries to keep many connections to the target web server open and hold
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

3. Weak Credential Auditing and Scope Adherence

The credential auditing phase utilizes the Hydra toolkit to perform parallel brute-force attacks. A critical aspect of this phase is the strict adherence to the project's "Weak Credentials" requirements (Criteria 2.2). The scanner programmatically filters the target list to include only **SSH, FTP, RDP, and TELNET**.

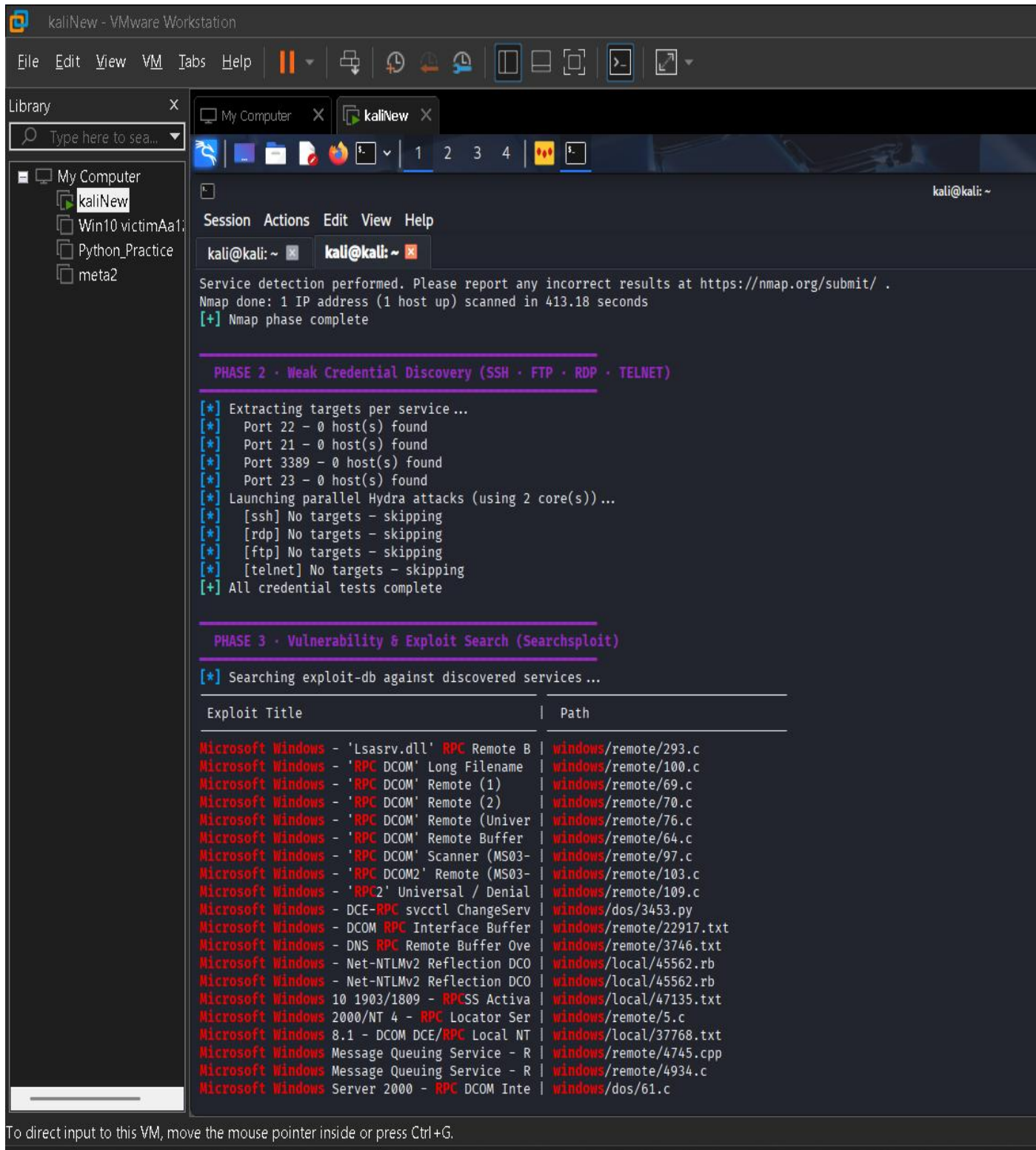
It is important to note that while **Port 445 (SMB) was detected** as open during the reconnaissance phase, it was **intentionally excluded** from the brute-force attack vector. This exclusion demonstrates precise compliance with the project specifications, which did not authorize brute-force testing against the SMB protocol. This logic ensures the tool operates strictly within the defined scope.



```
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File Edit View VM Tabs Help
Library
Type here to search
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Win10 victimAa1
Python_Practice
meta2
kali@kali: ~
Session Actions Edit View Help
kali@kali: ~
|
| them open as long as possible. It accomplishes this by opening connections to
| the target web server and sending a partial request. By doing so, it starves
| the http server's resources causing Denial Of Service.
|
| Disclosure date: 2009-09-17
| References:
|   https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|   http://ha.ckers.org/slowloris/
| _http-dombased-xss: Couldn't find any DOM based XSS.
| http-csrf:
| Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=172.20.10.2
| Found the following possible CSRF vulnerabilities:
|
|   Path: https://172.20.10.2:8089/static/atom.xml
|   Form id: controlpoint{position()}
|   Form action: {ahref}
| http-enum:
|   /robots.txt: Robots file
|   /services/: Potentially interesting folder (401 Unauthorized)
| _http-stored-xss: Couldn't find any stored XSS vulnerabilities.
| Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
| Device type: general purpose
| Running: Microsoft Windows XP|7|2012
| OS CPE: cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_server_2012
| OS details: Microsoft Windows XP SP3 or Windows 7 or Windows Server 2012
| Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
|
| Host script results:
| _smb-vuln-ms10-054: false
| _samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
| _smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|
| OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
| Nmap done: 1 IP address (1 host up) scanned in 383.95 seconds
| [*] Starting UDP scan (top 100 ports)...
| Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-18 17:21 EST
| Nmap scan report for 172.20.10.2
| Host is up (0.0013s latency).
| Not shown: 99 open/filtered udp ports (no-response)
| PORT      STATE SERVICE VERSION
| 49152/udp open  unknown
|
| 1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
|_SF-Port49152-UDP:V=7.95%I=7%D=2/18%T=69963C68%P=x86_64-pc-linux-gnu%r(S
|SF:qlping,1,"%x02");
|
| Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

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The auditing process supports both a built-in dictionary, generated securely via mktemp to prevent data leakage, and user-supplied password lists (Criteria 2.1). The execution is multi-threaded to ensure efficiency.



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the output of an Nmap scan and a Searchsploit session. The Nmap scan is complete, showing service detection performed. The Searchsploit session is in progress, displaying a list of exploit titles and their corresponding paths.

```
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Library
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My Computer
kaliNew
Win10 victimAa1
Python_Practice
meta2
Session Actions Edit View Help
kali@kali: ~ kali@kali: ~
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 413.18 seconds
[+] Nmap phase complete

PHASE 2 - Weak Credential Discovery (SSH - FTP - RDP - TELNET)

[*] Extracting targets per service ...
[*] Port 22 - 0 host(s) found
[*] Port 21 - 0 host(s) found
[*] Port 3389 - 0 host(s) found
[*] Port 23 - 0 host(s) found
[*] Launching parallel Hydra attacks (using 2 core(s)) ...
[*] [ssh] No targets - skipping
[*] [rdp] No targets - skipping
[*] [ftp] No targets - skipping
[*] [telnet] No targets - skipping
[+] All credential tests complete

PHASE 3 - Vulnerability & Exploit Search (Searchsploit)

[*] Searching exploit-db against discovered services ...

Exploit Title | Path
Microsoft Windows - 'Lsasrv.dll' RPC Remote B | windows/remote/293.c
Microsoft Windows - 'RPC DCOM' Long Filename | windows/remote/100.c
Microsoft Windows - 'RPC DCOM' Remote (1) | windows/remote/69.c
Microsoft Windows - 'RPC DCOM' Remote (2) | windows/remote/70.c
Microsoft Windows - 'RPC DCOM' Remote (Univer | windows/remote/76.c
Microsoft Windows - 'RPC DCOM' Remote Buffer | windows/remote/64.c
Microsoft Windows - 'RPC DCOM' Scanner (MS03- | windows/remote/97.c
Microsoft Windows - 'RPC DCOM2' Remote (MS03- | windows/remote/103.c
Microsoft Windows - 'RPC2' Universal / Denial | windows/remote/109.c
Microsoft Windows - DCE-RPC svcctl ChangeServ | windows/dos/3453.py
Microsoft Windows - DCOM RPC Interface Buffer | windows/remote/22917.txt
Microsoft Windows - DNS RPC Remote Buffer Ove | windows/remote/3746.txt
Microsoft Windows - Net-NTLMv2 Reflection DCO | windows/local/45562.rb
Microsoft Windows - Net-NTLMv2 Reflection DCO | windows/local/45562.rb
Microsoft Windows 10 1903/1809 - RPCSS Activa | windows/local/47135.txt
Microsoft Windows 2000/NT 4 - RPC Locator Ser | windows/remote/5.c
Microsoft Windows 8.1 - DCOM DCE/RPC Local NT | windows/local/37768.txt
Microsoft Windows Message Queuing Service - R | windows/remote/4745.cpp
Microsoft Windows Message Queuing Service - R | windows/remote/4934.c
Microsoft Windows Server 2000 - RPC DCOM Inte | windows/dos/61.c
```

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4. Vulnerability Mapping (NSE and Exploit-DB)

Since "Full Mode" was selected, the scanner initiates advanced vulnerability analysis. This includes the execution of Nmap Scripting Engine (NSE) scripts (--script vuln) to identify potential security flaws such as Slowloris or CSRF vulnerabilities directly from the service scan. Furthermore, the tool integrates Searchsploit to cross-reference the detected service versions (e.g., PostgreSQL 9.6, Windows RPC) against the Exploit-DB, providing a list of relevant exploits for potential escalation.

Library

Type here to sea...

My Computer

kaliNew

Win10 victimAa1

Python_Practice

meta2

My Computer

kaliNew

1234

Session

Actions

Edit

View

Help

kali@kali: ~

kali@kali: ~

Microsoft Windows Server 2000 SP4 - DNS RPC R | windows/remote/3737.py

Microsoft Windows XP/2000 - 'RPC DCOM' Remote | windows/remote/66.c

Microsoft Windows XP/2000 - RPC Remote Non Ex | windows/remote/117.c

Microsoft Windows XP/2000/NT 4.0 - RPC Servic | windows/dos/21951.c

Microsoft Windows XP/2000/NT 4.0 - RPC Servic | windows/dos/21952.c

Microsoft Windows XP/2000/NT 4.0 - RPC Servic | windows/dos/21953.txt

Microsoft Windows XP/2000/NT 4.0 - RPC Servic | windows/dos/21954.txt

Microsoft Windows XP/2003 - RPCSS Service Iso | windows/local/32892.txt

Shellcodes: No Results

Exploit Title	Path
VMware 2.5.1 - 'VMware-authd' Remote Denial o	multiple/dos/7647.txt
VMware GSX Server 2.0 - Authentication Server	windows/remote/21639.c
VMware Player / VMware Workstation 6.5.3 - 'V	windows/dos/33271.py
VMware vCenter Server 6.7 - Authentication By	multiple/webapps/48535.txt
VMware vCenter Server 7.0 - Remote Code Execu	multiple/webapps/50056.py
VMware vCenter Server 7.0 - Unauthenticated F	multiple/webapps/49602.py

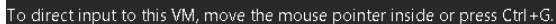
Shellcodes: No Results

Exploit Title	Path
PnPSCADA v2.x - Unauthenticated PostgreSQL In	hardware/webapps/51448.txt
PostgreSQL - 'bitsubstr' Buffer Overflow	linux/dos/33571.txt
PostgreSQL 6.3.2/6.5.3 - Cleartext Passwords	immunix/local/19875.txt
PostgreSQL 7.x - Multiple Vulnerabilities	linux/dos/25076.c
PostgreSQL 8.01 - Remote Reboot (Denial of Se	multiple/dos/946.c
PostgreSQL 8.2/8.3/8.4 - UDF for Command Exec	linux/local/7855.txt
PostgreSQL 8.3.6 - Conversion Encoding Remote	linux/dos/32849.txt
PostgreSQL 8.3.6 - Low Cost Function Informat	multiple/local/32847.txt
PostgreSQL 8.4.1 - JOIN Hashtable Size Intege	multiple/dos/33729.txt
PostgreSQL 9.3 - COPY FROM PROGRAM Command Ex	multiple/remote/46813.rb
PostgreSQL 9.3-11.7 - Remote Code Execution (	multiple/remote/50847.py
PostgreSQL 9.4-0.5.3 - Privilege Escalation	linux/local/45184.sh
PostgreSQL 9.6.1 - Remote Code Execution (RCE	multiple/remote/51247.py

Shellcodes: No Results

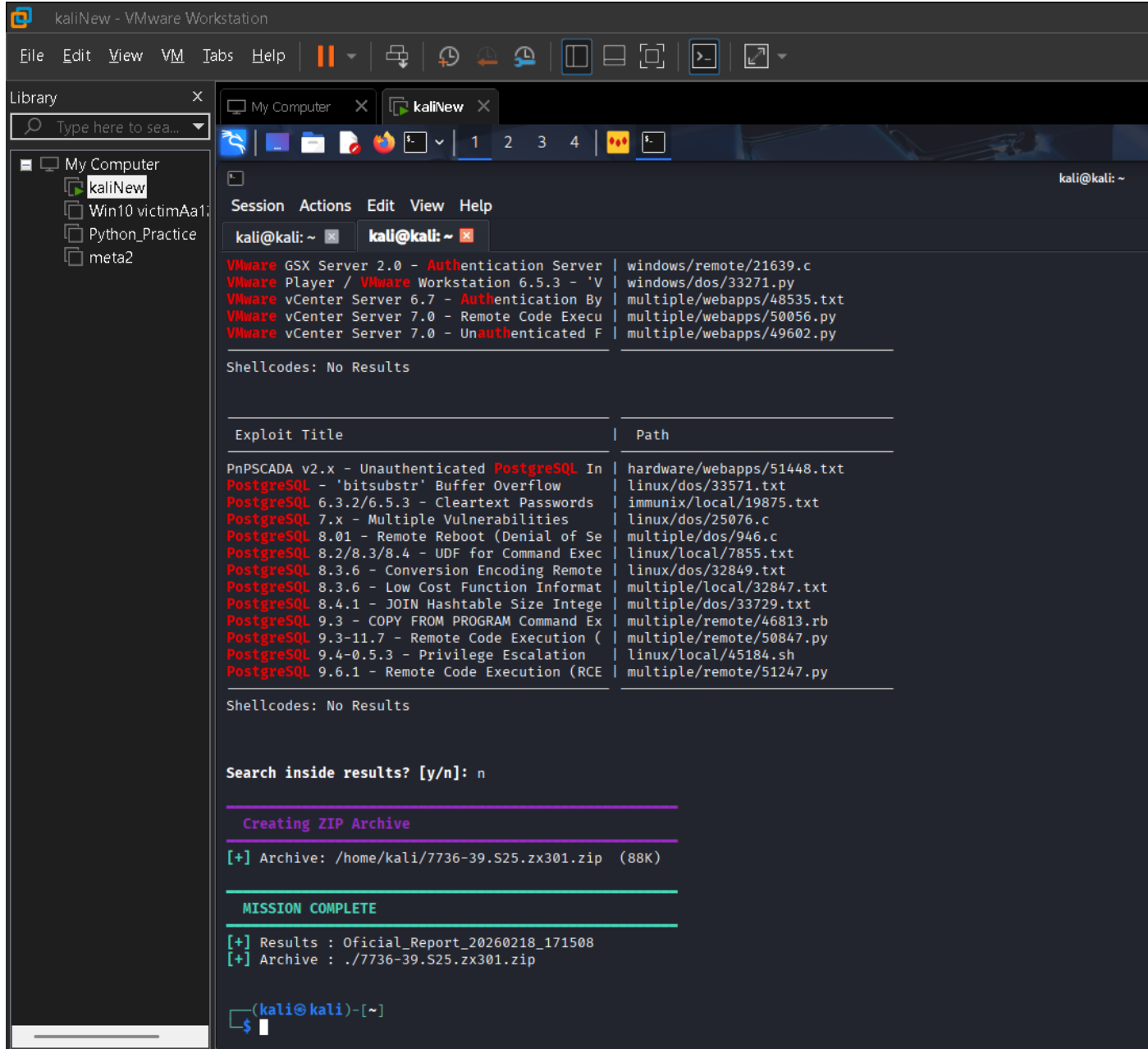
[!] Found 3 potential exploit(s)!

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



5. Reporting, Archiving, and Finalization

The final phase focuses on data persistence and usability. The scanner aggregates all findings into a structured directory and presents a summary of discovered hosts and services. An interactive search feature allows for immediate log analysis within the terminal. Finally, to meet Criteria 4.4, the entire workspace is compressed into a ZIP archive named according to the strict naming convention: UNIT.STUDENT.PROGRAM.zip.



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Win10 victimAa1
Python_Practice
meta2
Session Actions Edit View Help
kali@kali: ~ kali@kali: ~
VMware GSX Server 2.0 - Authentication Server | windows/remote/21639.c
VMware Player / VMware Workstation 6.5.3 - 'V | windows/dos/33271.py
VMware vCenter Server 6.7 - Authentication By | multiple/webapps/48535.txt
VMware vCenter Server 7.0 - Remote Code Execu | multiple/webapps/50056.py
VMware vCenter Server 7.0 - Unauthenticated F | multiple/webapps/49602.py
Shellcodes: No Results
Exploit Title | Path
PnPSCADA v2.x - Unauthenticated PostgreSQL In | hardware/webapps/51448.txt
PostgreSQL - 'bitsubstr' Buffer Overflow | linux/dos/33571.txt
PostgreSQL 6.3.2/6.5.3 - Cleartext Passwords | immunix/local/19875.txt
PostgreSQL 7.x - Multiple Vulnerabilities | linux/dos/25076.c
PostgreSQL 8.01 - Remote Reboot (Denial of Se | multiple/dos/946.c
PostgreSQL 8.2/8.3/8.4 - UDF for Command Exec | linux/local/7855.txt
PostgreSQL 8.3.6 - Conversion Encoding Remote | linux/dos/32849.txt
PostgreSQL 8.3.6 - Low Cost Function Informat | multiple/local/32847.txt
PostgreSQL 8.4.1 - JOIN Hashtable Size Intege | multiple/dos/33729.txt
PostgreSQL 9.3 - COPY FROM PROGRAM Command Ex | multiple/remote/46813.rb
PostgreSQL 9.3-11.7 - Remote Code Execution ( | multiple/remote/50847.py
PostgreSQL 9.4-0.5.3 - Privilege Escalation | linux/local/45184.sh
PostgreSQL 9.6.1 - Remote Code Execution (RCE | multiple/remote/51247.py
Shellcodes: No Results
Search inside results? [y/n]: n
Creating ZIP Archive
[+] Archive: /home/kali/7736-39.S25.zx301.zip (88K)
MISSION COMPLETE
[+] Results : Official_Report_20260218_171508
[+] Archive : ./7736-39.S25.zx301.zip
(kali@kali)-[~]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

